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Relation of white matter hyperintensities and motor deficits in chronic stroke

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Abstract.
Background: Infarct size and location account for only a relatively small portion of post-stroke motor impairment, suggesting that other less obvious factors may be involved.
Objective: Examine the relationship between white matter hyperintensity (WMH) load among other factors and upper extremity motor deficit in patients with mild to moderate chronic stroke.
Methods: The magnetic resonance images of 28 patients were studied. WMH load was assessed as total WMH volume and WMH overlap with the corticospinal tract in the centrum semiovale. Hemiparetic arm function was measured using the Motor Activity Log (MAL) and Wolf Motor Function Test (WMFT).
Results: Hierarchical multiple regression models found WMH volume predicted motor deficits in both real-world arm use (MAL; $\Delta R^2 = 0.12$, $F(1, 22) = 4.73$, $p = 0.04$) and in arm motor capacity as measured by a laboratory motor function test (WMFT; $\Delta R^2 = 0.18$, $F(1, 22) = 6.32$, $p = 0.02$) over and above age and lesion characteristics. However, these models accounted for less than half of the variance in post-stroke motor deficits.
Conclusion: The results suggest that WMH may be an important factor to consider in stroke-related upper extremity motor impairment. Nonetheless, the basis of the largest part of the post-stroke motor deficit remains unaccounted for by structural CNS factors. This component may be behavioral or learned, involving learned nonuse.

Keywords: Stroke, white matter hyperintensities, motor deficits, MRI, upper extremity, hemiparesis

1. Introduction

Motor deficit is a common consequence of stroke. However, its basis is unclear. Many studies of acute stroke report only weak to moderate relationships between lesion volume and motor deficits (Lin et al., 2017; Lövblad et al., 1997; Schiemanck, Post, Witkamp, Kappelle, & Prevo, 2005). Moreover, this relationship appears to decline over the first year post-stroke (Lauretani et al., 2010) and is mostly absent in the chronic phase (Mark, Taub, Perkins, Gauthier, & Uswatte, 2008; Page, Gauthier, & White, 2013). The only consistently replicated predictor of motor ability following chronic stroke is lesion overlap with the corticospinal tract (CST) (Gauthier et al., 2014; Rickards et al., 2014; Sterr, Dean, Szameitat, Conforto, & Shen, 2014; Zhu, Lindenberg, Alexander, & Schlaug, 2010), particularly in the centrum semiovale (CS) (Gauthier, Taub, Mark, Perkins, & Uswatte, 2009; Lo, Gitelman, Levy, Hulvershorn, & Parrish, 2010). Yet this still leaves a large majority of the variability of motor deficit in chronic stroke unaccounted for by lesion factors alone. Previous work from this laboratory found that there is a general decrease in

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amount of gray matter in non-infarcted motor areas associated with severity of the motor deficit (Gauthier, Taub, Mark, Barghi, & Uswatte, 2012), but this again accounted for only a portion of the motor deficit. Thus, neither the most obvious sources of motor deficit after stroke, the size and the location of the infarct, nor the general atrophy of the remaining cortical gray matter, account for a large proportion of motor impairment in chronic stroke. Consequently, other less obvious factors must be involved.

WMHs are brain abnormalities that appear as bright (i.e. "hyperintense") areas on proton-density, T2 weighted, and fluid attenuated inversion recovery (FLAIR) magnetic resonance imaging (MRI) scans and are presumably damaging due to their co-localization with microcystic infarcts, demyelination, and arteriosclerosis (Fazekas et al., 1993). Several studies show a relationship between WMHs and decreased cerebral blood flow or decreased cerebrovascular reactivity (Brickman et al., 2009; Marstrand et al., 2002), suggesting that WMHs may disturb neural communication. Many studies have associated WMHs with cognitive decline and aging (Kloppenborg, Nederkoorn, Geerlings, & van den Berg, 2014). Somewhat fewer studies have examined the relationship between WMHs and motor impairment (Linortner et al., 2012; Sachdev, Wen, Christensen, & Jorm, 2005) and most of these have been primarily with respect to lower extremity function such as balance and gait (Kreisel et al., 2013; Starr et al., 2003). While there have been some studies on WMHs and upper extremity motor function (Söderlund, Nyberg, Adolfsson, Nilsson, & Launer, 2003; Poggesi et al., 2013; Su et al., 2017), they did not involve separation of the effect of WMHs and stroke. In this study, we assessed in patients in the chronic phase after stroke with mild to moderate upper extremity motor deficit whether WMH load is related to the severity of the motor deficit. This study should help to clarify whether WMHs broadly disrupt neurologic function after stroke.

2. Methods

2.1. Participants

Chronic stroke patients with upper extremity hemiparesis were recruited for a study investigating the efficacy of full Constraint-Induced Movement therapy (CIMT) and a reduced form of the intervention (Taub et al., 2013). The majority of patients were self-referred (to obtain the treatment) and the remainder were referred by physicians associated with this urban institution. Twenty-eight of 40 candidate participants had pre-treatment MRI scans of sufficient quality and modality appropriate for this study. The scans did not reliably allow quantifying discrete strokes. Exclusion criteria included: post-stroke interval less than 10 months, considerable use of the more-affected arm (i.e., score greater than 2.5 out of 5 on the Motor Activity Log (MAL, described below), inability to follow basic verbal commands, use of pharmacological treatment for motor disability (e.g., botulinum toxin injections within the past 3 months), and history of having previously received CIMT. Inclusion criteria included ability to extend at least 20° at the wrist and 10° at each metacarpophalangeal joint (mild/moderate deficit) and absence of serious cognitive, visual, or auditory deficit or end-stage disease. The exclusion/inclusion criteria are described in greater detail elsewhere (Taub et al., 2013). Table 1 summarizes the characteristics of the participants. The pre-treatment MRI scans are presented in Fig. 1. It may be seen that WMHs were primarily localized in the area adjacent to the anterior and posterior horns of the lateral ventricles in both hemispheres, as well as in the superior periventricular space of the contralateral hemisphere. Of the patients' stroke lesions, eight patients had entirely subcortical lesions and the other 12 had stroke lesions that extended to the cerebral cortex. All participants provided signed informed consent, and the study was approved by our university's Institutional Review Board.

Table 1
Participant characteristics (N = 28)

	Mean (SD)
Age	63.2 (11.5)
Chronicity	3.27 (3.1)
MAL	1.3 (0.8)
WMFT	29.6 (13.9)
WMH Volume (cm^3)	21.5 (22.8)
WMH/CS Overlap (%): Ipsilateral	7.0 (10.3)
WMH/CS Overlap (%): Contralateral	7.4 (13.4)

Abbreviations: CS, centrum semiovale; MAL, Motor Activity Log; WMFT, Wolf Motor Function Test; WMH, white matter hyperintensity.

2.2. Measures of motor function

Two measures assessed motor function. The Wolf Motor Function Test (WMFT) is a validated laboratory measure of maximum motor capacity made on request (Morris, Uswatte, Crago, Cook, & Taub,

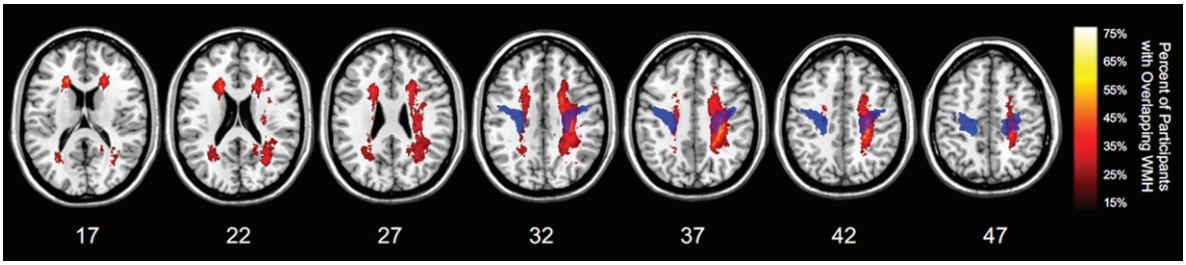


Fig. 1. Spatial distribution of WMHs. Voxels are colored (red-yellow) based on percent of participants with WMHs in that voxel. Brighter colors indicate greater percentage of participants with WMHs in that area. An ROI delineating the corticospinal tract in the centrum semiovale is overlaid (in blue). In hierarchical regression models, total WMH volume predicted motor deficits on both the MAL and the WMFT, but WMH overlap with the ROI did not (see text for details). The hemisphere contralateral to the more-affected arm is on the right. Numbers indicate MNI coordinates of each slice. Abbreviations: MAL, Motor Activity Log; ROI: region of interest; WMFT, Wolf Motor Function test; WMH, white matter hyperintensity.

2001; Wolf et al., 2001). It assesses performance time on 15 tasks (Taub et al., 1993); performance time was converted into a rate measure by calculating the rate of task performance over 60 seconds (60/average task completion time) (Hodics et al., 2012). The MAL is a scripted, structured, therapist-administered interview of how well and how often participants use their more-affected arm spontaneously for 30 commonly performed activities of daily living (Taub et al., 1993). The frequency (Amount of Use scale) and quality (Quality of Movement scale) of use of the more-affected upper extremity are rated according to scales with 6 steps (0 = not used and 5 = movement as good as pre-stroke), with half-step ratings allowed. Because the two scales of the MAL are highly correlated, only the Quality of Movement scale is reported and is heretofore referred to as the Arm Use Scale. This scale has been validated as a reliable measure of real-world arm use and correlates well with objective measures of arm use in the life situation (Uswatte et al., 2006; Uswatte, Taub, Morris, Light, & Thompson, 2006; Van der Lee, Beckerman, Knol, De Vet, & Bouter, 2004). The WMFT and the MAL measure different aspects of motor function: The WMFT measures ability to carry out standardized motor tasks upon request in the laboratory, while the MAL measures the actual amount of spontaneous, real-world arm use. The correlation between the two measures is only weak to moderate (Taub et al., 2006).

2.3. *MRI analysis*

FLAIR and T1 MRI scans were collected for all participants on a Phillips Intera 3 Tesla machine. T1 image acquisition parameters were axial 3D Turbo-Field Echo acquisition with 1 mm slice thickness, in-plane resolution of 0.94 mm × 0.94 mm, repetition time (TR) = 9.8, echo time (TE) = 4.6, field of view = 250, flip angle (α) = 8. FLAIR acquisition included 4 mm slices with 1 mm gaps, TR = 11,000, TE = 120, inversion time = 2,800, α = 90.

Hypointensive areas on T1 scans were interpreted as being cavitary lesions due to chronic stroke, regardless whether of ischemic or hemorrhagic etiology (Dewitt, Kistler, Miller, Richardson, & Buonanno, 1987; Hecht-Leavitt et al., 1986), and were manually traced independently by two investigators using a semi-automated thresholding method in MRIcron. WMHs were identified on FLAIR scans coregistered to the patients' T1 scans and were traced in the same manner as the lesions. To calculate inter-rater reliability of tracings, intraclass correlation coefficients (ICCs) were computed on a subset of 10 randomly selected subjects for both lesion (ICC = 0.97) and WMH (ICC = 0.91) tracings. All images were equated for side of motor deficit by flipping left to right the brains of patients with right side hemiparesis. Each patient's T1 scan was normalized to a standard T1 template in Montreal Neurological Institute (MNI) space using SPM8 (Wellcome Department of Cognitive Neurology, http://www.fil.ion.ucl.ac.uk/spm). These normalization parameters were then applied to each participant's coregistered FLAIR image, lesion tracing and WMH tracing. Lesion and WMH load were assessed first as total lesion or WMH volume, then as lesion or WMH overlap with the corticospinal tract in the centrum semiovale as a supplementary analysis. Lesion or WMH volume is reported as the total number of voxels identified in each image. A corticospinal tract (CST) region of interest (ROI) was created in standardized space using the SPM Anatomy Toolbox in SPM8 (Eickhoff et al., 2005). This ROI was then manually limited to the centrum semiovale (CS), which we defined as the

cerebral white matter superior to the lateral ventricles (MNI axial slice: +30 mm). We refer to this ROI as the CST in the CS. Lesion or WMH overlap with the CST in the CS is reported as the number of intersecting voxels divided by the total number of voxels in the CST in the CS ROI, expressed as a percentage. In previous work, the correlation between motor deficit and lesion overlap with the CST contralateral to a more-affected arm was found to be due predominantly to overlap with the CS, where input is received from the ipsilateral hemisphere over the corpus callosum (Gauthier et al., 2009).

2.4. Data analysis

Simple, bivariate correlations were examined between WMH volume, WMH overlap with the CST in the CS ipsilateral or contralateral to the more-affected upper extremity, lesion volume, lesion overlap with the CST in the CS ipsilateral or contralateral to the more-affected upper extremity, age, and the two measures of motor function. For the bivariate correlations, an $\alpha = 0.01$ threshold was used to control inflation of family-wise error.

To establish whether WMH volume or WMH overlap with the CST in the CS explains differences in motor function, after accounting for differences due to age and lesion characteristics, we used hierarchical multiple regressions. In the first regression model, age, and lesion volume were entered in the first block of predictors, lesion overlap with a) ipsilateral CST in the CS and b) contralateral CST in the CS were entered into the second block, and WMH volume was entered into the third and last block; the MAL score was the predicted variable. In the second model, the WMFT was substituted for the MAL. In addition, we tested parallel regression models that evaluated the predictive power of WMH overlap with the CST in the CS alone and WMH overlap with the CST in the CS after accounting for WMH volume. For the models that tested the predictive power of WMH overlap with the CST in the CS, the structure of the predictors was the same as in the previously described models except that WMH overlap with the CST in the CS was substituted for WMH volume in the last block of the model. For the models that tested the predictive power of WMH overlap with the CST in the CS after accounting for WMH volume, the structure of the predictors was again the same as the previous models except that WMH overlap with the CST in the CS was entered as a fourth block.

WMH volume in the CST and WMH volume in the CST of the CS were examined as predictors, in addition to whole-brain WMH, because lesion location studies in adults with chronic stroke suggest that damage to these regions has particular importance for upper-extremity motor deficit (e.g., Gauthier et al., 2009, 2014). MAL and WMFT scores were selected as predicted variables because they characterize separate, important domains of upper-extremity motor deficit (see *Measures of Motor Function*). We entered age and lesion volume into the first block of the regression models to permit testing of whether WMH volume accounted for variation in motor deficit of the more-affected arm over and above that accounted for by these two variables. Age was selected because previous studies have found that upper-extremity motor function declines with age (e.g., Desrosiers, Hebert, Bravo, & Rochette, 1998). Lesion volume was selected because (a) many neurologists hold the view that lesion volume influences motor function after stroke and (b) there is some support for this view in the acute phase (Lin et al., 2017; Lovblad et al., 1997; Schiemanch, Post, Witkamp, Kappelle, & Prevo, 2005), though this relationship appears to decline and then disappear as time since stroke increases (Lauretani et al., 2010; Mark et al., 2008; Page et al., 2013). Other variables that are thought by some to be related to post-stroke upper-extremity motor deficit, i.e., side of paresis, paresis of the pre-stroke dominant arm, chronicity, and race, were not entered into the first block because they have been found to be unrelated to pretreatment upper-extremity motor deficit or response to CI therapy previously (Taub et al., 2006, 2013).

3. Results

3.1. Participant characteristics

Table 1 summarizes the characteristics of the participants. Fig. 1 shows that WMHs were primarily localized in the area adjacent to the anterior and posterior horns of the lateral ventricles in both hemispheres, as well as in the superior periventricular space of the contralateral hemisphere.

3.2. Prediction of motor function by WMH volume

We first examined bivariate correlations between patient characteristics and motor scores. WMH

Table 2
Pearson correlations among selected participant characteristics and measures of motor function

	Age	MAL	WMFT	WMH Volume	WMH/CST Overlap: Ipsilateral	WMH/CST Overlap: Contralateral	Lesion Volume	Lesion/CST Overlap: Ipsilateral	Lesion/CST Overlap: Contralateral
Age	1	0.17	–0.08	–0.08	0.21	0.01	0.04	0.32	0.14
MAL		1	**0.69***	**–0.52***	–0.26	–0.17	–0.36	–0.23	–0.36
WMFT			1	**–0.53***	–0.39	–0.13	–0.14	–0.08	–0.41
WMH Volume				1	0.43	**0.55***	0.22	0.07	0.29
WMH/CST Overlap: Ipsilateral					1	–0.03	0.22	0.19	–0.01
WMH/CST Overlap: Contralateral						1	–0.12	–0.12	–0.06
Lesion Volume							1	**0.56***	0.12
Lesion/CST Overlap: Ipsilateral								1	–0.07
Lesion/CST Overlap: Contralateral									1

White matter hyperintensity volume is the only significant predictor of motor function. *Indicates significance at $p < 0.01$. Abbreviations: CST, corticospinal tract; MAL, Motor Activity Log; WMFT, Wolf Motor Function Test; WMH, white matter hyperintensity.

volume predicted motor function (MAL: $r = –0.52$, $p < 0.01$; WMFT: $r = –0.53$, $p < 0.01$), but no other participant characteristics did. Table 2 summarizes the results of these bivariate correlational analyses.

In the regression analyses, age, lesion volume, and lesion overlap with the CST in the CS did not predict motor function as measured by the WMFT ($R^2 = 0.18$, $F(4, 23) = 1.25$, $p = 0.32$). The amount of variance in MAL scores accounted for by these predictors also fell just short of significance ($R^2 = 0.32$, $F(4, 23) = 2.75$, $p = 0.052$). Adding WMH volume as a predictor resulted in a regression model that predicted spontaneous use of the more affected arm in everyday life situations as measured by the MAL more powerfully. This model as a whole accounted for a total of 44% of the variance in MAL scores ($R^2 = 0.44$, $F(5, 22) = 3.50$, $p = 0.02$), with WMH volume accounting for 12% of the variance in MAL scores ($\Delta R^2 = 0.12$, $F(1, 22) = 4.73$, $p = 0.04$) *over and above* that predicted by age, and lesion characteristics. When predicting motor function made on request in the laboratory as measured by the WMFT, the model trended toward significance. This model as a whole accounted for a total of 36% of the variance in WMFT scores ($R^2 = 0.36$, $F(5, 22) = 2.50$, $p = 0.06$), with WMH volume accounting for 18.3% of the variance in WMFT scores ($\Delta R^2 = 0.18$, $F(1, 22) = 6.32$, $p = 0.02$) *over and above* that of age and lesion characteristics. These regression analyses are summarized in Table 3.

Next, we examined the relationship between WMH overlap with the CST in the CS and motor ability. After accounting for age, lesion characteristics, and WMH volume, the addition of WMH overlap with the CST in the CS did not significantly improve the models (MAL: $\Delta R^2 = 0.02$, $F(2, 20) = 0.37$, $p = 0.70$; WMFT: $\Delta R^2 = 0.05$, $F(2, 20) = 0.76$, $p = 0.48$). Moreover, when WMH overlap with the CST in the CS was added *instead* of WMH volume, it did not account for a significant amount of variation in motor function as measured by the MAL ($\Delta R^2 = 0.13$, $F(2, 21) = 2.46$, $p = 0.11$) or the WMFT ($\Delta R^2 = 0.18$, $F(2, 21) = 2.91$, $p = 0.08$).

4. Discussion

The results show that, independent of age and lesion characteristics, WMH volume predicted the amount of upper-extremity motor deficit in patients with chronic stroke with deficits in the mild to moderate range. These results are consistent with previous findings that WMHs are related to poor motor function (Linortner et al., 2012; Sachdev et al., 2005).

In multiple regression analyses, age and lesion volume explained 16.5% of the variance in the motor deficit as measured by the MAL. Lesion overlap with the CST in the CS explained an additional 15.9% of the variance, and WMH volume explained a further 12.0% of the variance. For WMFT scores, age and lesion volume explained only 2.4% of the variance, while lesion overlap with the CST in the CS explained 15.5% of the variance and WMH volume explained an additional 18.3% of the variance. While WMH volume accounts for a significant portion of the variance over and above other factors, 56% to 64% of the motor deficit following stroke is still largely unexplained by the factors studied. To identify the source of this part of the motor deficit in chronic stroke, two possibilities seem worth considering: 1. behavioral factors including learned nonuse, 2. lesion

Table 3
Hierarchical multiple regression models

Measure	Model Block	Predictor Variable	β, t	R^2 Change
MAL	Block 1	Age	0.27, 1.53	0.17
		Lesion volume	–0.12, –0.58	
	Block 2	Lesion overlap with ipsilateral CST	–0.25, –1.19	0.16
		Lesion overlap with contralateral CST	–0.29, –1.68	
	Block 3	WMH volume	–0.37, –2.17*	0.12*
WMFT	Block 1	Age	–0.05, –0.28	0.02
		Lesion volume	0.03, 0.16*	
	Block 2	Lesion overlap with ipsilateral CST	–0.07, –0.29	0.16
		Lesion overlap with contralateral CST	–0.28, –1.49	
	Block 3	WMH volume	–0.46, –2.51*	0.18*

WMH volume predicts motor function as measured by the MAL and WMFT over and above age, and lesion characteristics. Age and lesion volume were entered into the model's first block. Lesion overlap with the CS in the ipsilateral hemisphere and lesion overlap with the CS in the contralateral hemisphere were entered into the model's second block. WMH volume was entered into the model's last block. *Indicates significance at $p \leq 0.05$. Abbreviations: CST, corticospinal tract; MAL, Motor Activity Log; WMFT, Wolf Motor Function Test; WMH, white matter hyperintensity.

and WMH-induced disruption of the operation of the brain networks that remain intact after the insult.

The role of behavioral factors in the motor deficits present after neurological injury is supported by both basic research in animals and clinical trials in humans. Monkeys whose somatic sensation had been surgically abolished by serial section of all dorsal roots of a single forelimb were observed to never spontaneously use that limb in the free colony environment (Mott & Sherrington, 1895). However, two behavioral techniques were found to restore extensive functional use of that limb: training of the deafferented extremity especially by the technique termed shaping, and continuous restraint of the intact forelimb for an extended duration (e.g., one week) (Taub, 1976). The restored movements were clumsy but nonetheless enabled walking, running, climbing, and even retrieving small food objects from shallow wells between thumb and forefinger. Thus, the largest part of the initial and normally permanent motor deficit could be overcome by behavioral techniques. The initial deficit was said to be due to a mechanism termed learned nonuse, whose operation was demonstrated in two experiments (Taub, 1976; Taub, 1980; Taub, Uswatte, Mark, & Morris, 2006). The same two behavioral techniques were later applied to humans to rehabilitate motor deficits after stroke and subsequently to a number of other types of damage to the CNS, including traumatic brain injury, cerebral palsy, multiple sclerosis and also to improve impaired speech in patients with non-fluent post-stroke aphasia (summarized in (Taub, Uswatte, & Mark, 2014)). The intervention was termed CI therapy. As noted, the techniques of CI therapy are largely behavioral. This in combination with the primate data suggests that a substantial part of the gross motor deficit after stroke may be behavioral or learned, involving the learned nonuse mechanism.

A second set of factors that could be responsible for the portion of post-stroke motor deficit whose basis is currently unaccounted for might be that lesions and WMHs, though not directly responsible for that moiety of the deficit per se, could have the effect of interfering with neural communication, whatever their location, through disrupting the operation of surviving brain networks. Even though they appear anatomically intact, their function could be seriously compromised by the structural damage produced by the insult and WMHs. Our findings are consistent with other studies indicating WMHs may interfere with neural communication (Brickman et al., 2009; Linortner et al., 2012; Marstrand et al., 2002; Reijmer et al., 2015). While lesions may eliminate one location in a network, the brain can compensate using intact tissue to substitute for the lost structural elements (Murphy & Corbett, 2009; Nudo, 2013; Taub et al., 2014). This may explain why many studies report weak to moderate relationships between lesion volume and motor deficits in stroke (Lauretani et al., 2010; Lövblad et al., 1997; Page et al., 2013; Parsons et al., 2000). WMHs on the other hand, which leave affected cortical regions partially intact, could result in abnormal signals being sent out from these disrupted regions that disturb neural communication more than infarcted tissue that has been destroyed. In the present study, we found that total WMH volume predicted motor deficit, but WMH overlap with the CST in the CS did not. This is unlike our

previous finding for lesion overlap with the CST in the CS which did predict motor deficit (Gauthier et al., 2009). Redundancy in the nervous system and gradually emerging post-event recovery mechanisms may provide at least partial compensation for neural tissue lost to a lesion (Murphy & Corbett, 2009); surviving but damaged tissue, however, may be a permanent source of error signals, which disturb effective brain system function. These behavioral and network factors are not mutually exclusive and could together be responsible for much of the motor deficit whose basis is presently unaccounted for.

While a significant relationship between lesion overlap with the CST in the CS and motor deficit in chronic stroke has been shown previously (Gauthier et al., 2014; Gauthier et al., 2009), the amount of WMH overlap with the CST in the CS did not predict motor scores over and above WMH volume in this study. This result may be due, in part, to the relatively consistent distribution of WMHs in and around the periventricular space near the CS in our subjects (see the Fig. 1), so that there is WMH overlap with the CST in the CS in most participants. Alternatively, the absence of a relationship between WMH overlap with the CST in the CS and motor impairment could suggest that WMH location is relatively unimportant and that presence of WMHs in locations other than the CS is enough to disrupt neural communication.

Although this research has reached its aim of assessing the relationship between WMH load and motor deficit in chronic stroke, there are several limitations in the present study. The sample size was relatively small ($n = 28$). The relationships established in the current study should be reexamined in a larger sample. In addition, the relation of WMHs to upper extremity motor deficit should be examined in less chronic patients and patients who have different levels of motor deficit severity to determine whether the relationships observed here in the chronic phase also pertain to the acute and subacute phases and to patients with less and more severe upper extremity motor deficits. Furthermore, to assess the degree to which WMHs may disturb underlying brain networks, follow-up studies should utilize EEG/MEG recording and functional MRI.

In spite of these limitations, the results suggest that WMH may be an important factor to consider in assessing the basis for upper extremity motor impairment in chronic stroke. These findings are consistent with previous data from this laboratory on: total lesion load, lesion overlap with the CST and CS, and interruption and displacement of the CST (Gauthier et al., 2009; Mark et al., 2008; Rickards et al., 2014). Finally, despite the relationship between WMH load and motor deficit, the majority of the post-stroke motor deficit remains unaccounted for.

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Disclosures

None.

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